

DATA VISUALIZATION · SUMMER 2026

Movies: Money, Ratings & Genres

What actually drives a film's success — budget, genre, timing, or acclaim?

10,178

films analysed

12

analytical questions

Live

Streamlit dashboard

Name: Sirisha Kummari Matriculation number: 72415466 Class:120B

IMDB Movies (Kaggle) — around 10,000 real films.

Why it was chosen

Real-world, rich, and varied — it mixes numbers, categories, time, and place, so it can be interrogated from many angles.

Columns used

rating · budget · revenue · genre · language · country · release date

Attribute types

Numerical, categorical, temporal and spatial — all four in one table.

Known caveats

The country field is skewed (Australia over-represented) and a few genre budgets look inflated — flagged in the analysis and treated as within-dataset.

10,178

films

7

key columns

4

attribute types

My approach: 12 analytical questions

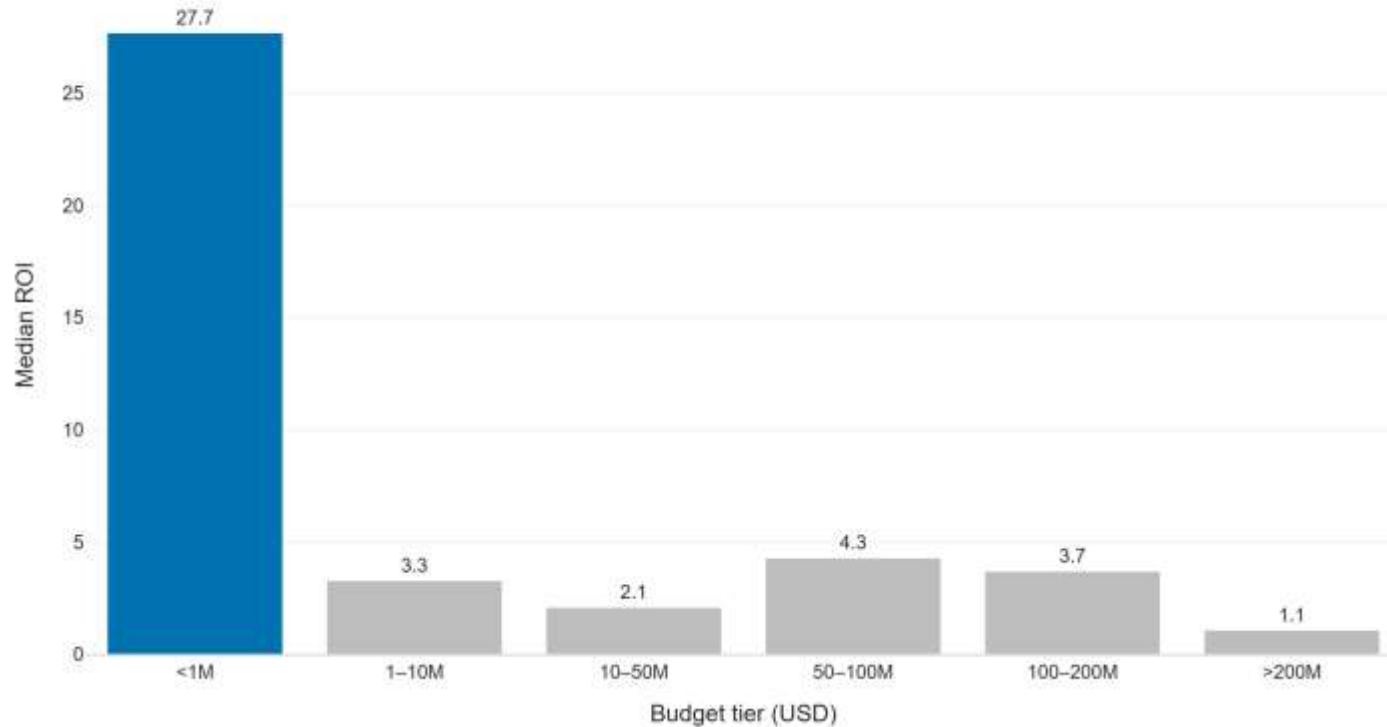
Each question is multi-dimensional and answered with one publication-ready Plotly chart.

- 1 Budget vs revenue
- 2 Genre return on budget (ROI)
- 3 Ratings across decades, by genre
- 4 English vs non-English: rating & budget
- 5 Countries punching above their weight
- 6 The budget “sweet spot” for ROI
- 7 Genre budgets vs revenue
- 8 Genre mix shifting over time
- 9 Critical acclaim vs box office
- 10 Best release months
- 11 Profitability by country
- 12 Rating–revenue link by budget tier

Budget & returns — the sweet spot

Lower-budget films tend to return the most per dollar spent

Median revenue + budget within each budget tier



Small budgets return the most per dollar.

Films made for under \$1M return about 27× their budget, while \$200M+ blockbusters barely break even. Efficiency falls steadily as budgets grow.

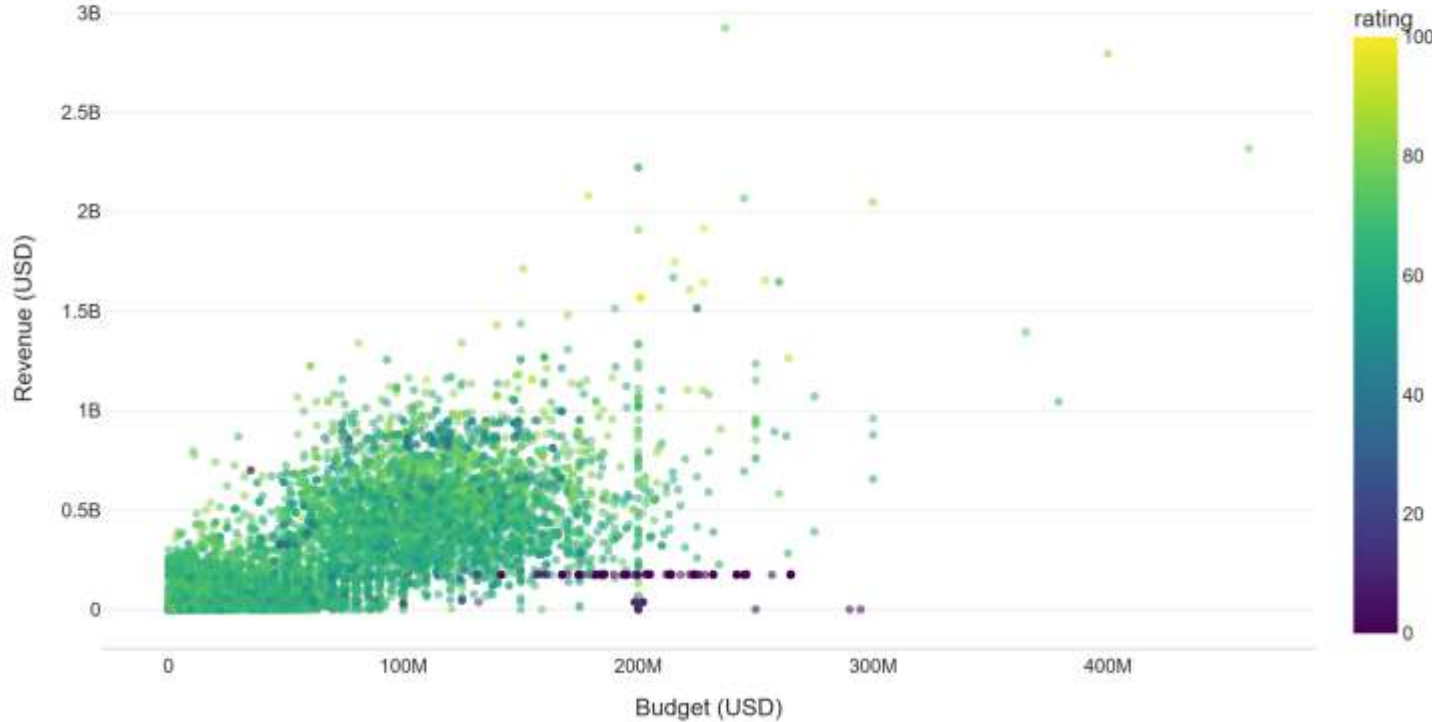
27.7×

median return, sub-\$1M films

Does spending pay off?

Bigger budgets tend to earn more — but the payoff is far from guaranteed

Each dot is a film - colour = rating



Bigger budgets usually earn more — but never guaranteed.

Budget and revenue rise together only moderately. The wide spread shows money alone doesn't buy a hit.

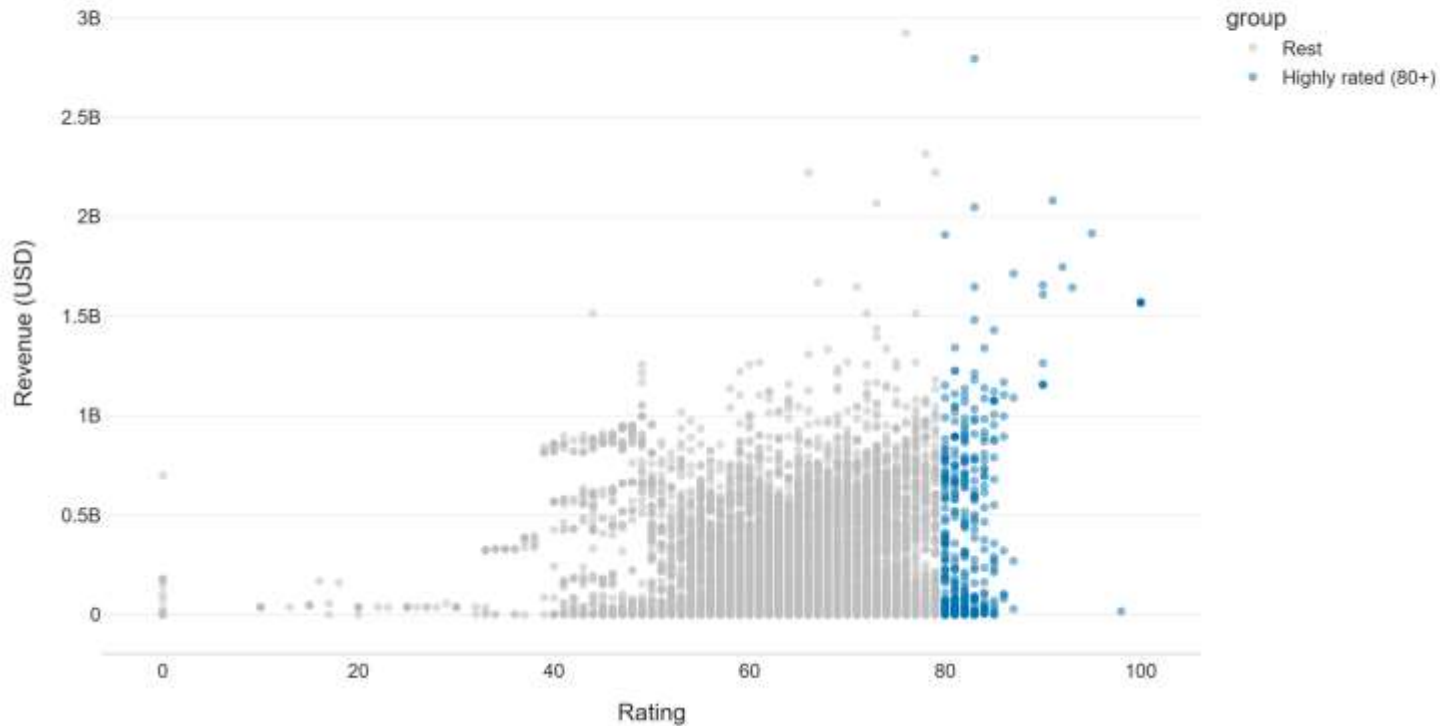
$$r \approx 0.67$$

budget–revenue correlation

Acclaim vs box office

High ratings don't guarantee high box office

Each dot is a film



A great film isn't necessarily a profitable one.

Across all films, rating and revenue are almost unrelated. Highly-rated titles are scattered across every revenue level.

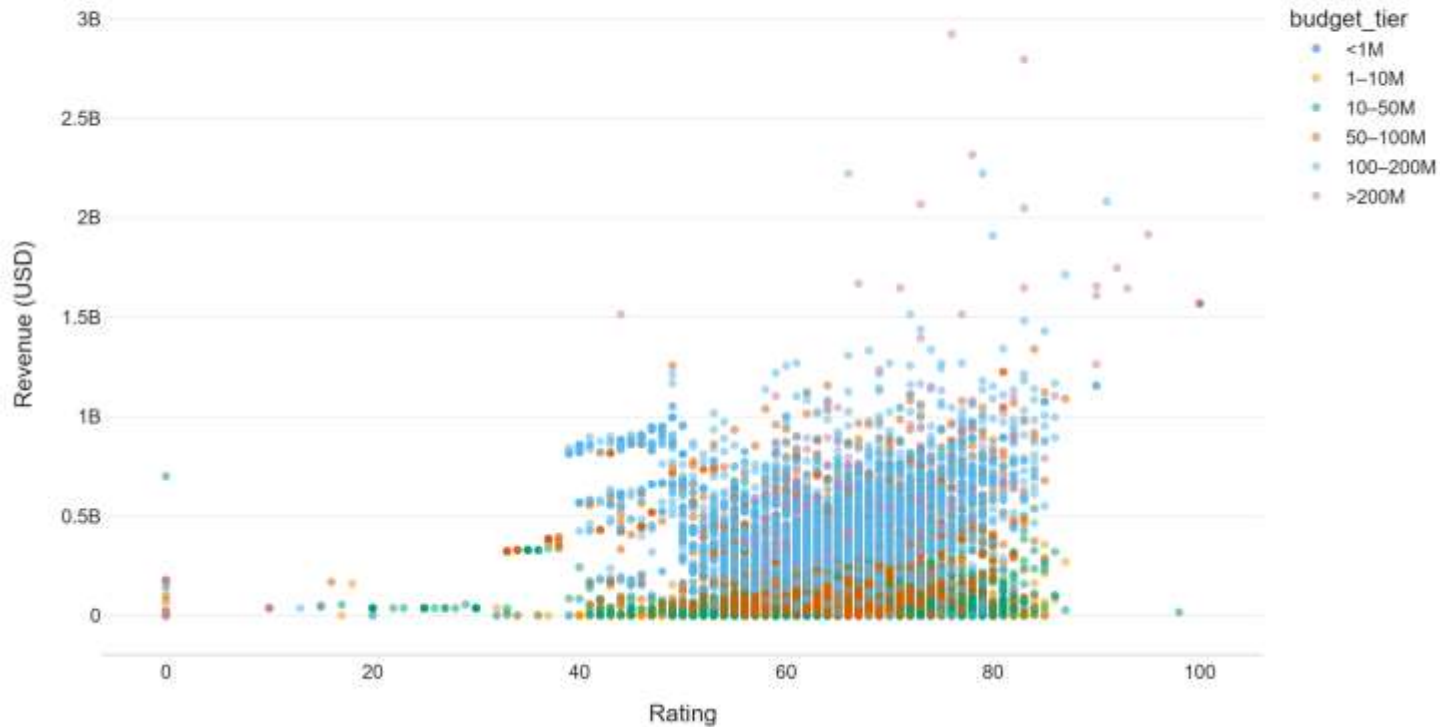
$$r \approx 0.10$$

rating–revenue correlation

Acclaim, conditioned on budget

Budget shapes the rating–revenue link

Each dot is a film - colour = budget tier



**Quality converts to money
mainly for big-budget films.**

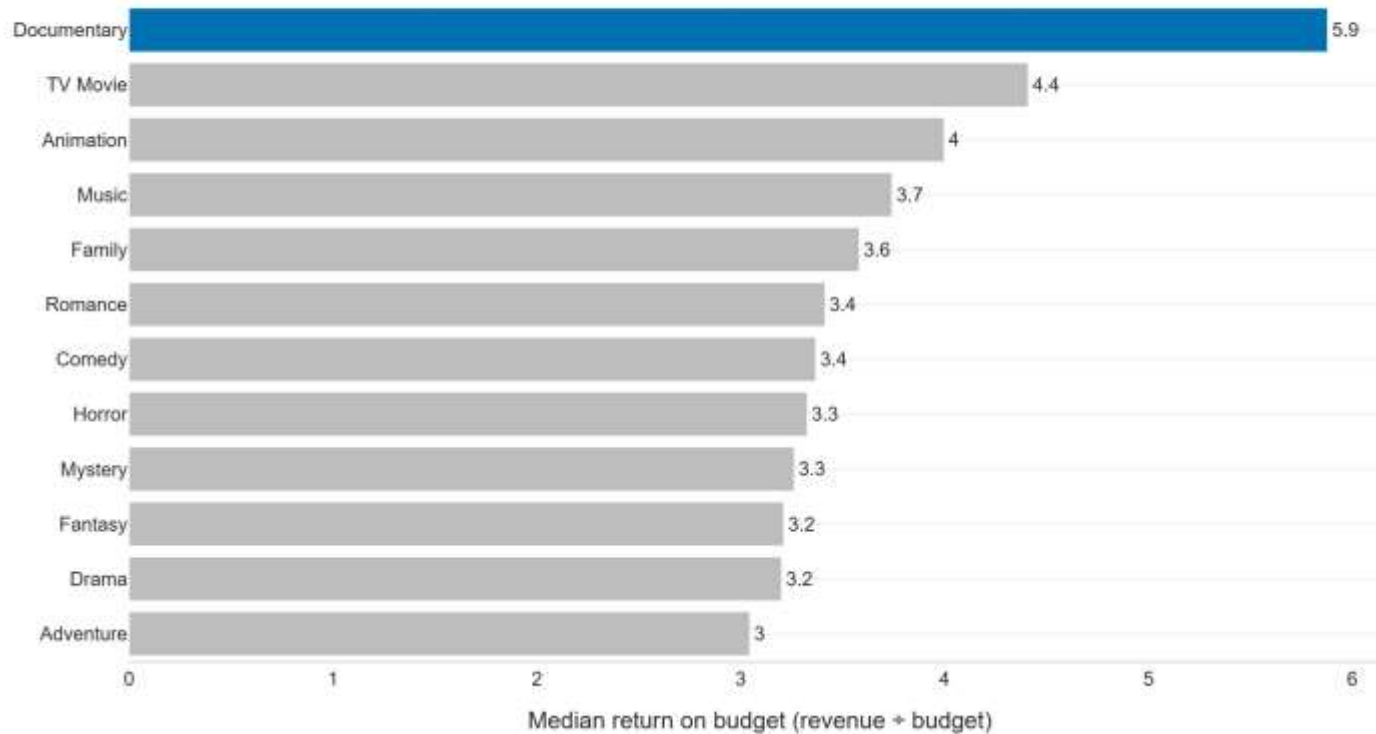
The rating–revenue link is near zero for cheap films but grows strong for the biggest ones. For blockbusters, being good is what earns.

0.14 → 0.79

correlation, cheap → \$200M+ films

Documentary films give the strongest return on budget

Median revenue per \$1 of budget, by genre



Lean genres win on return per dollar.

Documentary, TV Movie and Animation top the ROI table — modest budgets that punch well above their weight.

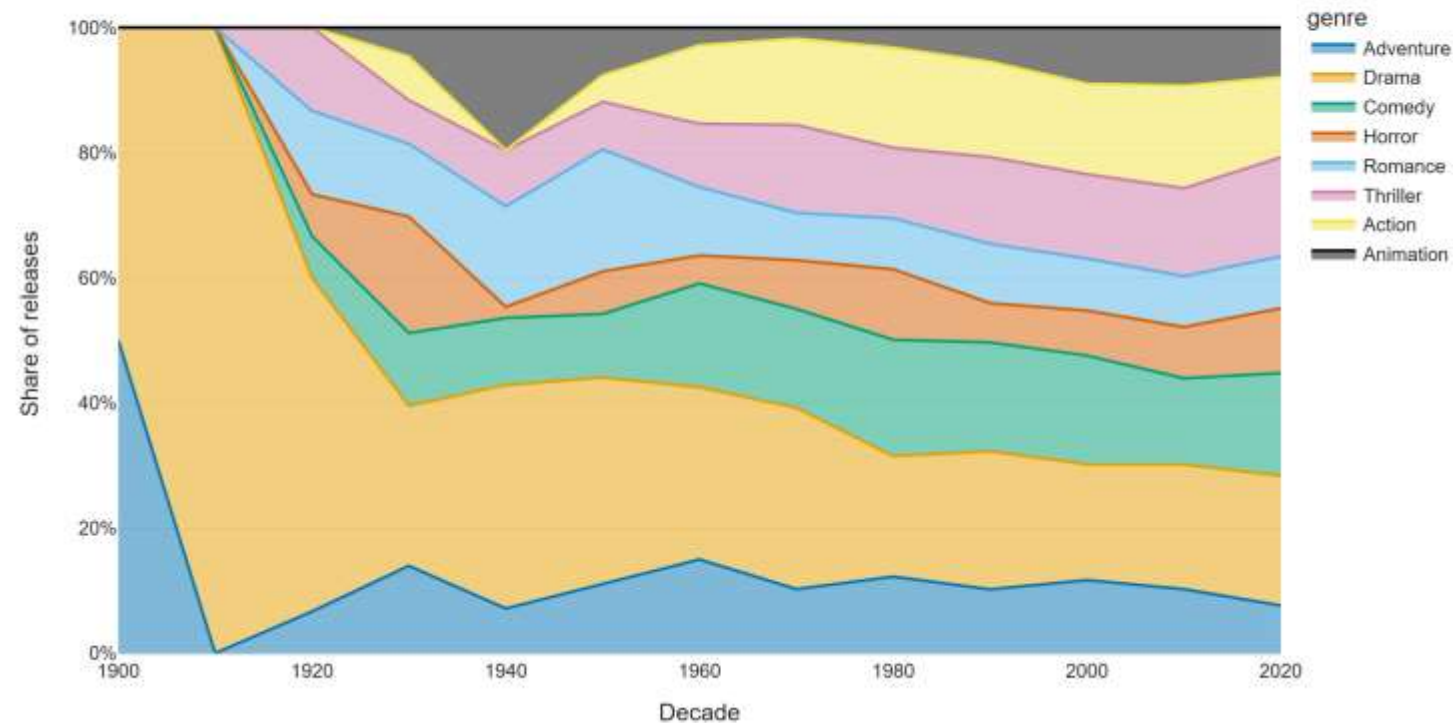
5.9x

median ROI, Documentary

Genres over time

The genre balance of releases has shifted over the decades

Share of films by genre within each decade



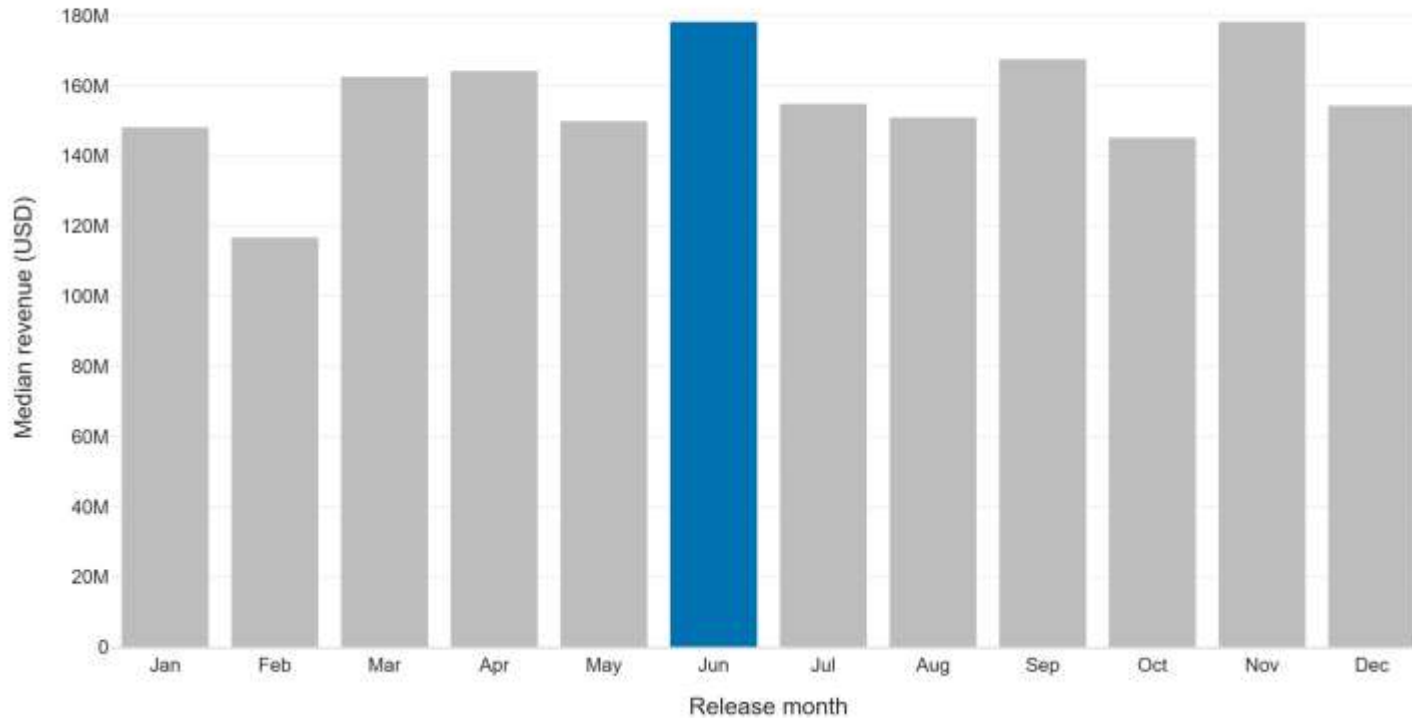
Cinema diversified from a Drama / Adventure core.

Early releases were dominated by two genres. Today the mix spreads broadly across comedy, thriller, horror and action.

Release timing

Films released in peak season out-earn the rest

Median box-office revenue by release month



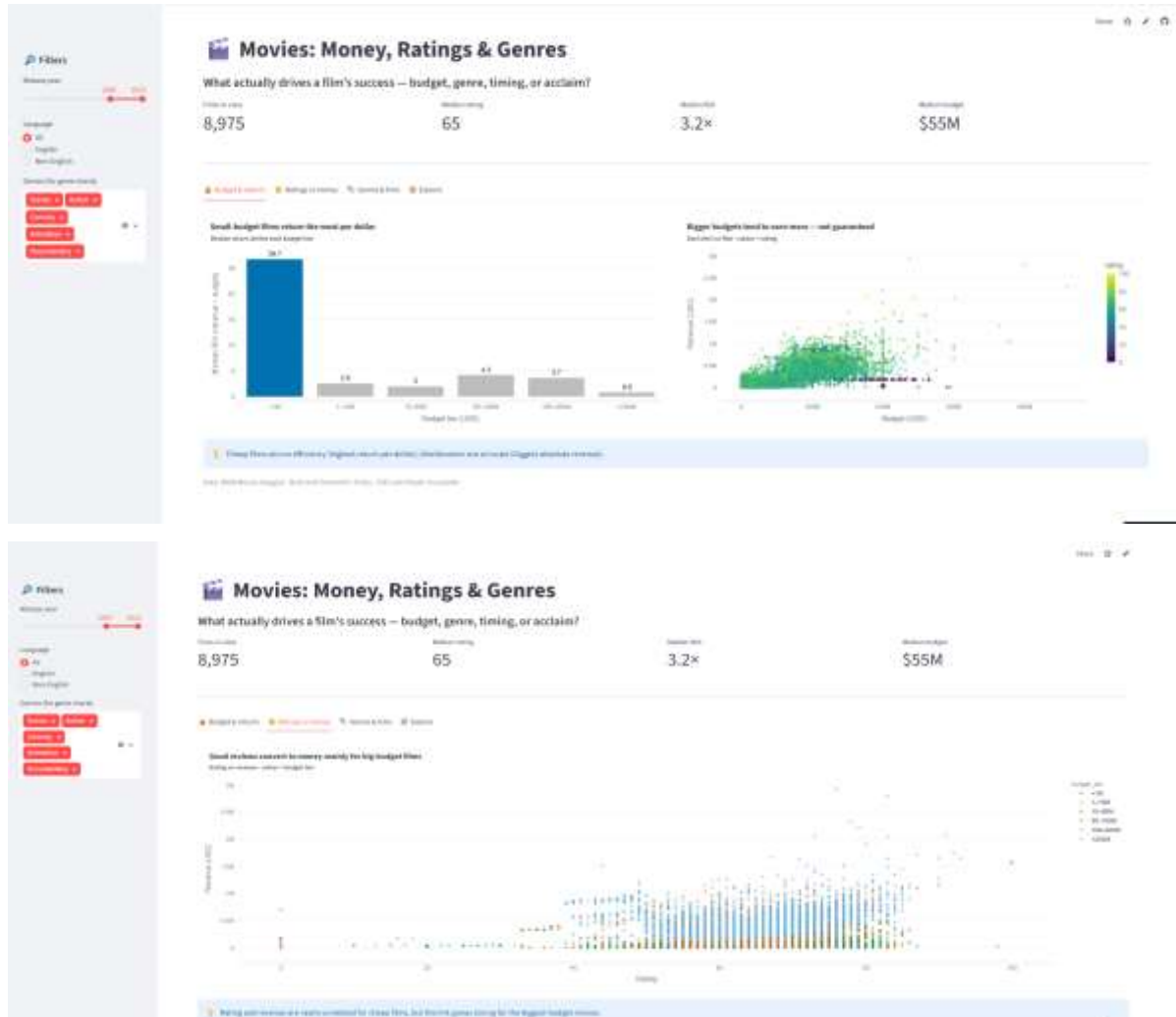
Peak-season releases out-earn the rest.

June and November post the highest median box office (~\$178M); February is weakest. Timing clearly tilts the odds.

Jun & Nov

highest-earning release months

Interactive dashboard



Live & explorable

- Key metrics update as you filter
- Year, language and genre selectors
- Four themed tabs of curated charts
- Same CVD-safe palette throughout

Live dashboard

<https://imdb-movie-analytics-uqxfgls6f7fgcofjeczubu.streamlit.app/>

GitHub repository

<https://github.com/Sirisha-kummari/IMDb-Movie-Analytics>

The big picture

Budget fundamentally changes how a film succeeds.

Cheap films

Win on efficiency — the highest return per dollar, almost regardless of reviews.

Big films

Win on scale — the largest revenue, and the only tier where quality reliably turns into money.

Acclaim alone

Barely predicts earnings. Genre and timing tilt the odds; the genre mix keeps broadening.

Cheap films compete on return, expensive films compete on quality-at-scale — and a good rating alone was never enough.

Thank you

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